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Modular lighting system assembly

Abstract

A modular lighting system includes a base that includes three cable connectors, a first cable connected to a first cable connector and connecting the base to a first adjacent base, a second cable connected to a second cable connector and connecting the base to a second adjacent base, a third cable connected to a third cable connector and connecting the base to a lighting unit, and the lighting unit. The first adjacent base is disposed upstream from the base, and the second adjacent base is disposed downstream from the base. The first, second, and third cables are interchangeable such that they can connect to any of the three cable connectors. Power transmitted along the third cable is less than power transmitted along the first cable and than power transmitted along the second cable.

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Background/Summary

RELATED APPLICATIONS (1) This application is a continuation of U.S. non-provisional patent application Ser. No. 17/960,721, filed Oct. 5, 2022 to Nikos Aristotle LAMBESIS et al., titled “MODULAR LIGHTING SYSTEM ASSEMBLY,” the entirety of the disclosure of which is hereby incorporated by reference.

TECHNICAL FIELD

(1) Aspects of this document relate generally to modular lighting systems and more specifically to improving power supply to lights farther from the main power source.

BACKGROUND

(2) Both outdoor spaces (such as backyards, walkways, and patios) and indoor spaces (such as restaurants, clubs, and other venues) can often be enhanced by increasing the beauty, appeal, and utility of the spaces with the addition of elements such as lighting. Some conventional lighting systems provide a range of colors and intensities, allowing for the creation of a lighting setup tailored to a particular space or a particular function or activity, such as a party. However, bespoke conventional lighting systems can be very expensive, requiring careful installation and wiring. Such systems, though expensive and custom made, often lack features such as individually addressable lights. Once installed, rearranging the lighting units can be time consuming and expensive, sometimes requiring replacement of a portion of the system. While less expensive conventional lighting systems exist, they have their own set of drawbacks. The cost of such systems is low because they can be mass produced, with lights spaced evenly along a line, such as every two feet. Such a constraint can make such systems difficult to adapt to locations needing variable spacing between the lights.

(3) In addition, even where a modular lighting system is used, there may be some limits to the design. For example, modular systems may face difficulties in providing enough power to lights farther down a line of the lighting system away from the main power source. Thus, modular lighting systems that improve power supply to lights farther away from a power source are desirable.

SUMMARY

(4) Aspects of this document relate to a modular lighting system. These aspects may comprise, and implementations may include, one or more or all of the components and steps set forth in the appended CLAIMS.

(5) In one aspect, a modular lighting system includes a base that has three cable connectors, a first cable connected to a first cable connector of the three cable connectors, a second cable connected to a second cable connector of the three cable connectors, a third cable connected to a third cable connector of the three cable connectors, and a lighting unit. In some embodiments, the first cable connects the base to a first adjacent base in a series of bases, with the first adjacent base being disposed upstream from the base. In some embodiments, the second cable connects the base to a second adjacent base in the series of bases, with the second adjacent base being disposed downstream from the base. In some embodiments, the third cable connects the base to the lighting unit. In some embodiments, the first cable, the second cable, and the third cable are interchangeable such that the first cable, the second cable, and the third cable can connect to any of the first cable connector, the second cable connector, and the third cable connector. In some embodiments, a power transmitted along the third cable from the base to the lighting unit is less than a power transmitted along the first cable from the first adjacent base to the base. In some embodiments, the power transmitted along the third cable is less than a power transmitted along the second cable from the base to the second adjacent base.

(6) In some embodiments, the first cable connector and the second cable connector are disposed on opposite sides of the base from each other. In some embodiments, the third cable connector is disposed at a 90-degree angle with respect to the first cable connector and the second cable connector. In some embodiments, the third cable connector is disposed on a bottom of the base. In some embodiments, the lighting unit is disposed below the base. In some embodiments, the base is a T-shaped connection. In some embodiments, the lighting unit includes one or more light emitting diodes.

(7) In one aspect, a modular lighting system includes a base that includes three cable connectors, a

base circuit board, and a microcontroller mounted on the base circuit board. In some embodiments, the system includes a first cable connected to a first cable connector of the three cable connectors, a second cable connected to a second cable connector of the three cable connectors, a third cable connected to a third cable connector of the three cable connectors, and a lighting unit that includes a lighting unit circuit board and one or more light emitting diodes mounted on the lighting unit circuit board. In some embodiments, the first cable connects the base to a first adjacent base in a series of bases, with the first adjacent base being disposed upstream from the base. In some embodiments, the second cable connects the base to a second adjacent base in the series of bases, with the second adjacent base being disposed downstream from the base. In some embodiments, the third cable connects the base to the lighting unit.

(8) In some embodiments, the microcontroller mounted on the base circuit board controls the light emitting diodes mounted on the lighting unit circuit board. In some embodiments, an amount of power transmitted along the third cable is less than an amount of power transmitted along the first cable and an amount of power transmitted along the second cable. In some embodiments, the first cable connector and the second cable connector are disposed on opposite sides of the base from each other. In some embodiments, the third cable connector is disposed at a 90-degree angle with respect to the first cable connector and the second cable connector. In some embodiments, the third cable connector is disposed on a bottom of the base. In some embodiments, the lighting unit is disposed below the base.

(9) In one aspect, a modular lighting system includes a control unit, a plurality of bases arranged in series, with a first base of the plurality of bases being connected to the control unit, a plurality of lighting units, with each lighting unit being connected to one of the plurality of bases, and a plurality of cables including a first cable connecting the control unit and the first base of the plurality of bases, a set of base cables, with each base cable connecting two bases of the plurality of bases, and a set of lighting unit cables, with each lighting unit cable connecting one of the plurality of lighting units to one of the plurality of bases. In some embodiments, the first cable, the base cables, and the lighting unit cables are interchangeable.

(10) In some embodiments, the first cable, the base cables, and the lighting unit cables include cables of different lengths. In some embodiments, the base cables include cables of different lengths. In some embodiments, the lighting unit cables include cables of different lengths. In some embodiments, an amount of power transmitted along the lighting unit cables is less than an amount of power transmitted along the first cable and the base cables. In some embodiments, each of the plurality of lighting units is disposed below the base to which the lighting unit is connected.

(11) In one aspect, a modular lighting system includes a control unit, a plurality of bases arranged in series, with a first base of the plurality of bases being connected to the control unit, a plurality of lighting units, with each lighting unit being connected to one of the plurality of bases, and a plurality of cables including a first cable connecting the control unit and the first base of the plurality of bases, a set of base cables, with each base cable connecting two bases of the plurality of bases, and a set of lighting unit cables, with each lighting unit cable connecting one of the plurality of lighting units to one of the plurality of bases.

(12) In some embodiments, the first cable and the base cables are interchangeable. In some embodiments, the lighting unit cables are different than the first cable and the base cables such that the lighting unit cables are not interchangeable with the first cable or the base cables. In some embodiments, the lighting unit cables are stiffer than the first cable and the base cables. In some embodiments, a diameter of the lighting unit cables is different than a diameter of the first cable and a diameter of the base cables. In some embodiments, the diameter of the lighting unit cables is less than the diameter of the first cable and the diameter of the base cables. In some embodiments, an amount of power transmitted along the lighting unit cables is less than an amount of power transmitted along the first cable and the base cables. In some embodiments, each base includes a base circuit board and a microcontroller mounted on the base circuit board. In some embodiments,

each lighting unit includes a lighting unit circuit board and one or more light emitting diodes mounted on the lighting unit circuit board.
(13) The foregoing and other aspects, features, and advantages will be apparent to those artisans of ordinary skill in the art from the DESCRIPTION and DRAWINGS, and from the CLAIMS.
(14) Unless specifically noted, it is intended that the words and phrases in the specification and the claims be given their plain, ordinary, and accustomed meaning to those of ordinary skill in the applicable arts. The inventors are fully aware that they can be their own lexicographers if desired. The inventors expressly elect, as their own lexicographers, to use only the plain and ordinary meaning of terms in the specification and claims unless they clearly state otherwise and then further, expressly set forth the “special” definition of that term and explain how it differs from the plain and ordinary meaning. Absent such clear statements of intent to apply a “special” definition, it is the inventors' intent and desire that the simple, plain and ordinary meaning to the terms be applied to the interpretation of the specification and claims.
(15) The inventors are also aware of the normal precepts of English grammar. Thus, if a noun, term, or phrase is intended to be further characterized, specified, or narrowed in some way, then such noun, term, or phrase will expressly include additional adjectives, descriptive terms, or other modifiers in accordance with the normal precepts of English grammar. Absent the use of such adjectives, descriptive terms, or modifiers, it is the intent that such nouns, terms, or phrases be given their plain, and ordinary English meaning to those skilled in the applicable arts as set forth above.
(16) Further, the inventors are fully informed of the standards and application of the special provisions of 35 U.S.C. § 112(f). Thus, the use of the words “function,” “means” or “step” in the Detailed Description or Description of the Drawings or claims is not intended to somehow indicate a desire to invoke the special provisions of 35 U.S.C. § 112(f), to define the invention. To the contrary, if the provisions of 35 U.S.C. § 112(f) are sought to be invoked to define the inventions, the claims will specifically and expressly state the exact phrases “means for” or “step for”, and will also recite the word “function” (i.e., will state “means for performing the function of [insert function]”), without also reciting in such phrases any structure, material or act in support of the function. Thus, even when the claims recite a “means for performing the function of . . . ” or “step for performing the function of . . . ,” if the claims also recite any structure, material or acts in support of that means or step, or that perform the recited function, then it is the clear intention of the inventors not to invoke the provisions of 35 U.S.C. § 112(f). Moreover, even if the provisions of 35 U.S.C. § 112(f) are invoked to define the claimed aspects, it is intended that these aspects not be limited only to the specific structure, material or acts that are described in the preferred embodiments, but in addition, include any and all structures, materials or acts that perform the claimed function as described in alternative embodiments or forms of the disclosure, or that are well known present or later-developed, equivalent structures, material or acts for performing the claimed function.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The disclosure will hereinafter be described in conjunction with the appended drawings, where like designations denote like elements, and:
(2) FIG. **1** is a perspective view of a modular lighting system according to some embodiments;
(3) FIG. **2** is a perspective view of a base, a cable, and a lighting unit that may be used in a modular lighting system according to some embodiments;
(4) FIG. **3** is a cross-section view of the base, cable, and lighting unit of FIG. **2** according to some embodiments;

(5) FIG. **4** is a perspective view of a base, a cable, and a lighting unit that may be used in a modular lighting system according to some embodiments; and
(6) FIG. **5** is a cross-section view of the base, cable, and lighting unit of FIG. **4** according to some embodiments.
DETAILED DESCRIPTION
(7) This disclosure, its aspects and implementations, are not limited to the specific material types, components, methods, or other examples disclosed herein. Many additional material types, components, methods, and procedures known in the art are contemplated for use with particular implementations from this disclosure. Accordingly, for example, although particular implementations are disclosed, such implementations and implementing components may comprise any components, models, types, materials, versions, quantities, and/or the like as is known in the art for such systems and implementing components, consistent with the intended operation.
(8) The word "exemplary," "example," or various forms thereof are used herein to mean serving as an example, instance, or illustration. Any aspect or design described herein as "exemplary" or as an "example" is not necessarily to be construed as preferred or advantageous over other aspects or designs. Furthermore, examples are provided solely for purposes of clarity and understanding and are not meant to limit or restrict the disclosed subject matter or relevant portions of this disclosure in any manner. It is to be appreciated that a myriad of additional or alternate examples of varying scope could have been presented but have been omitted for purposes of brevity.
(9) While this disclosure includes a number of embodiments in many different forms, there is shown in the drawings and will herein be described in detail particular embodiments with the understanding that the present disclosure is to be considered as an exemplification of the principles of the disclosed methods and systems and is not intended to limit the broad aspect of the disclosed concepts to the embodiments illustrated.
(10) Both outdoor spaces (such as backyards, walkways, and patios) and indoor spaces (such as restaurants, clubs, and other venues) can often be enhanced by increasing the beauty, appeal, and utility of the spaces with the addition of elements such as lighting. Some conventional lighting systems provide a range of colors and intensities, allowing for the creation of a lighting setup tailored to a particular space or a particular function or activity, such as a party. However, bespoke conventional lighting systems can be very expensive, requiring careful installation and wiring. Such systems, though expensive and custom made, often lack features such as individually addressable lights. Once installed, rearranging the lighting units can be time consuming and expensive, sometimes requiring replacement of a portion of the system. While less expensive conventional lighting systems exist, they have their own set of drawbacks. The cost of such systems is low because they can be mass produced, with lights spaced evenly along a line, such as every two feet. Such a constraint can make such systems difficult to adapt to locations needing variable spacing between the lights.
(11) In addition, even where a modular lighting system is used, there may be some limits to the design. For example, modular systems may face difficulties in providing enough power to lights farther down a line of the lighting system away from the main power source. Thus, modular lighting systems that improve power supply to lights farther away from a power source are desirable, thus providing more flexibility in the design of the modular system.
(12) For example, it may be desirable to use lights that hang from a ceiling in an indoor space or to use lights that are low to, but spaced from, the ground in an outdoor space. Running power along cables in such a system may use up too much power so that the lights farther from the power source do not have enough power. Contemplated herein is a lighting system that improves power supply for lights farther from the power source, while maintaining modularity such that lighting units are able to be rearranged at will and deployed in any desired configuration. In some embodiments, the contemplated modular lighting system comprises a control unit, a plurality of cables, a plurality of lighting units, and a base associated with each lighting unit. The cables may connect the control

unit with a first base, adjacent bases to each other, and bases with their respective lighting units. In some embodiments, the cables are interchangeable with each other. In some embodiments, at least one of the cables is not interchangeable with other cables. For example, in some embodiments, cables connecting lighting units to bases may not be interchangeable with cable connecting the control unit with a base and cables connecting adjacent bases to each other.
(13) In some embodiments, less power is transmitted along the cables connecting a base with a lighting unit than the cables connecting adjacent bases. Each base may comprise a circuit board with a controller mounted on the circuit board. Each lighting unit may comprise a circuit board with one or more light-emitting diodes mounted on the circuit board. By using two circuit boards, with one at the base and one at the lighting unit, less power may be transmitted along the cables connecting a base with a lighting unit to provide power to the lighting unit. This shortens the distance of cable along which full power is being transmitted, thus improving the power supply to lights farther away from the power source. The modular lighting systems disclosed herein may use any of the features disclosed in U.S. application Ser. No. 16/929,668, filed Jul. 15, 2020, U.S. application Ser. No. 17/728,851, filed Apr. 25, 2022, and U.S. application Ser. No. 17/867,626, filed Jul. 18, 2022. The disclosures of these three applications are hereby incorporated by reference in their entireties into this disclosure.
(14) A modular lighting system is shown, for example, in FIG. **1** as modular lighting system **100**. In some embodiments, modular lighting system **100** comprises one or more spaced lighting arrangements **102** (each including a base **120**, a lighting unit **130**, and a cable **108** connecting the base **120** to the lighting unit **130**), a control unit **104**, and a plurality of cables **108** connecting the control unit **104** and the one or more spaced lighting arrangements **102**. The one or more spaced lighting arrangements **102** are arranged in series.
(15) In some embodiments, control unit **104** is configured to control lighting units **130** of spaced lighting arrangements **102**. The control unit **104** may provide power and instructions to spaced lighting arrangements **102**, causing lighting units **130** to emit light **110**. In some embodiments, control unit **104** comprises a user interface **106**. User interface **106** may allow a user to perform various operations with respect to modular lighting system **100**, such as turning the modular lighting system **100** on and off, selecting a mode for modular lighting system **100** (e.g., selecting a color of light emitted, selecting a pattern of how light is emitted, etc.), dimming the lights, and performing other operations. In some embodiments, control unit **104** and/or user interface **106** may have any of the features of the control units and user interfaces described in U.S. application Ser. No. 16/929,668, filed Jul. 15, 2020, U.S. application Ser. No. 17/728,851, filed Apr. 25, 2022, and U.S. application Ser. No. 17/867,626, filed Jul. 18, 2022.
(16) In some embodiments, cables **108** convey the power and instructions from control unit **104** to the spaced lighting arrangements **102**. Cables **108** may be provided in a variety of lengths so that different arrangements of system **100** may be made depending on the space in which lighting system **100** is being installed. The ends of each cable **108** may be configured to connect with control unit **104**, bases **120**, or lighting units **130**. For example, ends of cables **108** may be provided with threads to mate with cable connectors on the control unit **104**, bases **120**, and lighting units **130**. Other types of attachments may alternatively be used instead of threads. In some embodiments, the ends of cable **108** are identical to each other so that the orientation of cable **108** is not important, which may facilitate quicker assembly of system **100**. Cables **108** may include any of the features of the cables described in U.S. application Ser. No. 16/929,668, filed Jul. 15, 2020, U.S. application Ser. No. 17/728,851, filed Apr. 25, 2022, and U.S. application Ser. No. 17/867,626, filed Jul. 18, 2022.
(17) The reference numeral **108** is used generally to refer to any cable **108** used in system **100**. Other reference numerals are also used to refer to various cables to specify which cable is being referred to. For example, a cable **108** between control unit **104** and a first base **120** may be referred to as a cable **103**, a cable **108** between two bases **120** may be referred to as a cable **105**, and a cable

108 between a base **120** and a lighting unit **130** may be referred to as a cable **107**. As another example, in discussing a spaced lighting arrangement **102** and its adjacent cables **108**, a cable **108** connecting the base **120** of that spaced lighting arrangement **102** and an adjacent base **120** upstream from that spaced lighting arrangement **102** (i.e., closer to control unit **104**) may be referred to as a cable **122**, a cable **108** connecting the base **120** to the lighting unit **130** of that spaced lighting arrangement **102** may be referred to as a cable **124**, and a cable **108** connecting the base **120** of that spaced lighting arrangement **102** and an adjacent base **120** downstream from that spaced lighting arrangement **102** (i.e., farther away from control unit **104**) may be referred to as a cable **126**.

(18) It should be understood, however, that these other reference numerals are just for convenience in describing system **100**. In some embodiments, cables **108** are interchangeable such that any cable **108** may be used to connect any two components to allow for modularity of system **100**. Thus, the other reference numerals for cables **108** (e.g., **103**, **105**, **107**, **122**, **124**, **126**) designate a location of a cable **108** with respect to other components in system **100** and not necessarily a difference in structure. For example, a cable **108** may be arranged between a control unit **104** and the first base **120** (i.e., be a cable **103**) in one arrangement and subsequently be arranged between two bases **120** (i.e., be a cable **105**) or between a base **120** and a lighting unit **130** (i.e., be a cable **107**) in another arrangement. Even in one arrangement, a cable **108** may be a cable **126** with respect to one base **120** (i.e., be disposed downstream to connect to an adjacent base **120**) and be a cable **122** with respect to that adjacent base **120** (i.e., be disposed upstream from that adjacent base **120**).

(19) In some embodiments, not all cables **108** are interchangeable. For example, in some embodiments, cable **103** and cables **105** may be interchangeable while cables **107** between a lighting unit **130** and a base **120** are not interchangeable with cable **103** or cables **105**. In some embodiments, cables **107** are different than the first cable and the base cables such that the lighting unit cables are not interchangeable with the first cable or the base cables. For example, cables **107** may be stiffer than cable **103** and cables **105**. As another example, a diameter of the cables **107** may be different than a diameter of the cable **103** and a diameter of cables **105**. In some embodiments, the diameter of the cables **107** may be less than the diameter of cable **103** and the diameter of cables **105**. In some embodiments, this difference may be for aesthetic reasons, such as to provide a sleek design for the cables **107** extending down to lighting units **130**. In some embodiments, this difference may be due to a difference in requirements for the cables. For example, cables **107** may have fewer wires (e.g., two or three wires) than the number of wires in cable **103** and cables **105** (e.g., five or six wires) because less communication and less power is required to be transmitted from bases **120** to lighting units **130** when compared to communication and power being transmitted from control unit **104** to bases **120** and from one base **120** to another base **120**. Other examples of cables **108** not being interchangeable are possible in other embodiments.

(20) In some embodiments, spaced lighting arrangement **102** comprises a base **120**, a lighting unit **130**, and a cable **108** connecting the base **120** to the lighting unit **130** (i.e., a cable **107** or **124**). Lighting unit **130** is configured to emit light **110** when system **100** is powered and running (depending on instructions from control unit **104**). Additional details about spaced lighting arrangements (such as lighting arrangement **102**) are provided below with respect to FIGS. **2-5**. In some embodiments, lighting unit **130** (or base **120**) may have any of the features of the lighting units described in U.S. application Ser. No. 16/929,668, filed Jul. 15, 2020, U.S. application Ser. No. 17/728,851, filed Apr. 25, 2022, and U.S. application Ser. No. 17/867,626, filed Jul. 18, 2022.

(21) As illustrated in FIG. **1**, modular lighting system **100** includes, in some embodiments, control unit **104**, a plurality of bases **120** arranged in series (with a first base **120** connected to control unit **104**), a plurality of lighting units **130**, and a plurality of cables **108**. In some embodiments, each lighting unit **130** is connected to one of the plurality of bases **120**. The plurality of cables **108** may

include a first cable **103** connecting the control unit **104** and the first base **120**, a set of base cables **105**, and a set of lighting unit cables **107**. Each base cable **105** connects two bases **120** to each other. Each lighting unit cable **107** connects one of the lighting units **130** to one of the bases **120**. In some embodiments, the first cable **103**, the base cables **105**, and the lighting unit cables **107** are interchangeable. Thus, as noted above, the designation as the first cable **103**, a base cable **105**, and a lighting unit cable **107** indicates the arrangement of those cables **108** rather than necessarily a structural difference (e.g., first cable **103** may subsequently be a base cable **105** or a lighting unit cable **107** in a different arrangement of system **100**).

(22) In some embodiments, first cable **103**, base cables **105**, and lighting unit cables **107** include cables of different lengths. For example, at least one of these cables **103**, **105**, **107** may have a different length than another one of the cables **103**, **105**, **107**. In some embodiments, each cable **103**, **105**, **107** may have a unique length, different from any other cable **103**, **105**, **107**. In other embodiments, each cable **103**, **105**, **107** has the same length. In some embodiments, cable **103** may have a different length than cables **105** (which may all have the same length), and cables **107** (which may all have the same length) may have a different length than cable **103** and cables **105**.

(23) In some embodiments, base cables **105** include cables of different lengths. For example, at least one of base cables **105** may have a different length than another one of base cables **105**. In some embodiments, each base cable **105** may have a unique length, different from any other base cable **105**. In other embodiments, each base cable **105** has the same length. In some embodiments, lighting unit cables **107** include cables of different lengths. For example, at least one of lighting unit cables **107** may have a different length than another one of lighting unit cables **107**. In some embodiments, each lighting unit cable **107** may have a unique length, different from any other lighting unit cable **107**. In other embodiment, each lighting unit cable **107** has the same length. In some embodiments, an amount of power transmitted along the lighting unit cables **107** is less than an amount of power transmitted along the first cable **103** and the base cables **105**. As discussed in more detail below, sending less power along lighting unit cables **107** may help supply power to lighting units **130** that are farther away from the power source (e.g., control unit **104**).

(24) In some embodiments, as shown in FIG. **1**, the lighting units **130** are disposed below bases **120** (i.e., lighting units **130** are arranged such that they are closer to a ground surface or floor than their respective bases **120**). For example, each lighting unit **130** in FIG. **1** is disposed below the base **120** to which the lighting unit **130** is connected. This arrangement may be used, for example, when the lighting units **130** are spaced from the ceiling. For example, bases **120** may be connected to the ceiling and lighting units **130** may be disposed below the bases **120** (e.g., hanging by lighting unit cable **107**). In some embodiments, as discussed below and shown in FIGS. **4** and **5**, lighting units **130** may be disposed above the bases **120** (i.e., lighting units **130** are arranged such that they are farther away from a ground surface or floor than their respective bases **120**). This arrangement may be used, for example, when the lighting units **130** are spaced from the ground. For example, bases **120** may be connected to, placed on, or inserted into the ground and lighting units **130** may be disposed above the bases **120**.

(25) Modular lighting system **100** is illustrated in FIG. **1** to show various components of system **100** and how they fit together. The illustrated system **100** is, however, merely representative and does not limit how the system **100** may be arranged. For example, although four lighting units **130** are shown, more or fewer lighting units **130** may be used as part of system **100**. Because lighting system **100** is modular, any number of variations may be made to how the system **100** is arranged to fit a specific indoor or outer space. These variations may include, but are not limited to, the length of the cable **103** between the control unit **104** and the first base **120**, the lengths of the cables **105** between bases **120** (which may all be the same length, all be different lengths, or be a mixture of being the same and different lengths), the lengths of the cables **107** between each base **120** and its respective lighting unit **130** (which may all be the same length, all be different lengths, or be a mixture of being the same and different lengths), and the number of bases **120** and lighting units

130.

(26) Thus, a user may assemble the system **100** to fit a particular space by selecting where to place the control unit **104**, selecting where to place the lighting units **130** within the space (e.g., exact locations where lighting units **130** are desired, how far to space each lighting unit **130** from adjacent lighting units **130**, how far to space each lighting unit **130** from the ceiling or ground, etc.), selecting cables **108** of appropriate lengths so that each lighting unit **130** is in its desired location, and then assembling the components and installing the components in the space. Assembling the components and installing the components in the space may be done in any order. For example, the bases **120** and control unit **104** may be installed first and then all cables **108** and lighting units **130** attached. As another example, the entire system **100** may be assembled first and then installed in the space. As yet another example, the system **100** may be assembled and installed in parallel (e.g., the control unit **104** is installed, a cable **103** is attached to the control unit **104**, a base **120** is attached to the other end of cable **103**, the base **120** is installed (e.g., coupled to the ceiling or installed in the ground), a cable **107** is attached to the base **120**, a lighting unit **130** is attached to the other end of cable **107**, a cable **105** is attached to the base **120**, another base **120** is attached to the other end of cable **105**, etc.). Other variations in assembling and installing the system **100** may also be used.

(27) An example of spaced lighting arrangement **102** (comprising base **120**, lighting unit **130**, and cable **108** connecting the base **120** and the lighting unit **130**) is shown in more detail in FIGS. **2** and **3**. In some embodiments, base **120** comprises three cable connectors, including a first cable connector **121**, a second cable connector **123**, and a third cable connector **125**. Each cable connector **121**, **123**, **125** is configured to connect to a cable **108**. For example, cable connectors **121**, **123**, **125** may be threaded or have some other mating feature corresponding to an end of cables **108**. Cable connectors **121**, **123**, **125** may have any of the features of the cable connectors disclosed in U.S. application Ser. No. 16/929,668, filed Jul. 15, 2020, U.S. application Ser. No. 17/728,851, filed Apr. 25, 2022, and U.S. application Ser. No. 17/867,626, filed Jul. 18, 2022.

(28) In some embodiments, a first cable **122** is connected to first cable connector **121** (see FIG. **1** for the cable **122** and FIGS. **2** and **3** for the cable connector **121**). First cable **122** may connect base **120** to a first adjacent base **120** disposed upstream in a series of bases **120**. In some embodiments, a second cable **126** is connected to second cable connector **123** (see FIG. **1** for the cable **126** and FIGS. **2** and **3** for the cable connector **123**). Second cable **126** may connect base **120** to a second adjacent base **120** disposed downstream in a series of bases **120**. In some embodiments, a third cable **124** is connected to third cable connector **125** (see FIGS. **1-3** for the cable **124** and FIGS. **2** and **3** for the cable connector **125**). Third cable **124** may connect base **120** to lighting unit **130**. As discussed above, cables **122**, **124**, **126**, in some embodiments, are interchangeable such that they are configured to connect to any of first cable connector **121**, second cable connector **123**, and third cable connector **125**. In some embodiments, cable **124** may not be interchangeable with cables **122** or **126**, as discussed above with respect to cable **107** and cables **103** and **105**.

(29) In some embodiments, first cable connector **121** and second cable connector **123** are disposed on opposite sides of base **120** from each other, as shown in FIG. **2**. First cable connector **121** and second cable connector **123** may be aligned with each other such that they share a common axis. In other embodiments, first cable connector **121** and second cable connector **123** are disposed at angles with respect to each other. For example, first cable connector **121** and second cable connector **123** may be disposed at right angles to each other (i.e., 90-degree angles), which may facilitate a change in direction for a string of spaced lighting arrangements **102**. In some embodiments, base **120** may have additional cable connectors, allowing a user to select which cable connector to use when assembling system **100**.

(30) In some embodiments, third cable connector **125** is disposed at a right angle (i.e., 90-degree angle) with respect to first cable connector **121** and second cable connector **123**. First cable connector **121**, second cable connector **123**, and third cable connector **125** may have their axes in a

single plane. In some embodiments, third cable connector **125** is disposed on a bottom of the base **120** (e.g., the portion of the base disposed closest to the ground or floor and farthest from the ceiling), as shown in FIG. **2**. Lighting unit **130** may be disposed below base **120**. With the first cable connector **121**, second cable connector **123**, and third cable connector **125** arranged as shown in FIGS. **2** and **3**, base **120** may be a T-shaped connection between cables **122**, **124**, **126**.
(31) In some embodiments, cable **108** (which may also be referred to as cable **107** or cable **124**) that connects lighting unit **130** to base **120** (via third cable connector **125**) comprises two ends **128**. One end **128** is connected to third cable connector **125** and the other end **128** is connected to a cable connector **135** (see FIG. **3**) of lighting unit **130**. Ends **128** may be the same, allowing cable **108** to be installed in either direction, furthering the modular nature of system **100**. Ends **128** may include threads or some other mating feature to correspond with cable connectors **125** and **135** (and other cable connectors of system **100**). To facilitate this, cable connectors **121**, **123**, **125**, **135** may each have the same shape and size. A gap is shown in cable **108** to represent that cables **108** of different lengths may be used as the cable **124** that connects a base **120** and a lighting unit **130**.
(32) As shown in FIG. **3**, base **120** may comprise a base circuit board **140**. In some embodiments, base circuit board **140** is disposed within base **120** (e.g., within a housing of base **120**). Base **120** may provide electrical connections between cable connectors **121**, **123**, **125** and base circuit board **140** so that power and communications from control unit **104** will reach base circuit board **140** and be able to be transmitted on to either lighting unit **130** or the next base **120**. In some embodiments, a microcontroller **142** is mounted on base circuit board **140**.
(33) As also shown in FIG. **3**, lighting unit **130** may comprise a lighting unit circuit board **150**. In some embodiments, lighting unit circuit board **150** is disposed within lighting unit **130** (e.g., within a housing of lighting unit **130**). Lighting unit **130** may provide electrical connections between cable connector **135** and lighting unit circuit board **150** so that power will reach lighting unit circuit board **150**. In some embodiments, one or more light emitting diodes **152** are mounted on lighting unit circuit board **150**.
(34) Microcontroller **142** may be configured to control light emitting diodes **152**. For example, microcontroller **142** may receive communication from control unit **104** and, based on that communication, light up light emitting diodes **152** so that lighting unit **130** emits light according to the communication from control unit **104**. By placing microcontroller **142** in base **120** (rather than in lighting unit **130**), less power can be transmitted along cable **124** than is transmitted along cables **122**, **126**. For example, power to light up the light emitting diodes **152** is all that is needed. Additional power to run the microcontroller **142** and send more complex commands is not needed through cable **124**. Thus, in some embodiments, power transmitted along cable **124** from base **120** to lighting unit **130** is less than power transmitted along cable **122** from an adjacent upstream base **120** to that particular base **120**. The power transmitted along cable **124** from base **120** to lighting unit **130** may also be less than power transmitted along cable **126** from that particular base **120** to an adjacent downstream base **120**. Because less power is being transmitted along the cables **124** connecting a base **120** with a lighting unit **130** to provide power to the lighting unit **130**, the total distance of cables **108** along which full power is being transmitted is shorter, thus improving the power supply to lighting units **130** farther away from the power source (e.g., control unit **104**).
(35) Because less power is being transmitted along cable **124**, cable **124** may be different than cables **122** and **126** such that cables **124** are not interchangeable with cables **122** and **126**, as discussed above for cable **107** with respect to cables **103** and **105**. For example, cable **124** may have a smaller diameter and/or contain fewer wires. In some embodiments, cable **124** may have a larger diameter and/or be stiffer than cables **122** and **126**. However, in other embodiments, cable **124** is interchangeable with cables **122** and **126** (e.g., same diameter, same number of wires, same stiffness) to have greater modularity and simplify the setup of system **100**. Having different lengths does not make the cables non-interchangeable.
(36) As mentioned above, lighting system **100** may be used to space lights from the ceiling or the

ground. For example, base **120** may be coupled to a ceiling and lighting unit **130** be spaced from the ceiling, hanging from base **120** via cable **108**. In some embodiments, base **120** may be coupled to a ceiling via bolts, screws, or other fasteners. In some embodiments, base **120** may be coupled to a ceiling via adhesives. Base **120** may be coupled to a ceiling via other fastening means (e.g., hook and loop fasteners, etc.). Base **120** may be coupled to a ceiling via a combination of the disclosed techniques, or any other technique. In some embodiments, base **120** itself may include a component configured to couple to the ceiling (e.g., a protruding component configured to pierce a ceiling, etc.).

(37) In some embodiments, base **120** may instead be coupled to a ground with any of the techniques described above, or others. In some embodiments, cable **124** may be stiff enough that it can support lighting unit **130** to be spaced from the ground. In some embodiments, cable **124** is stiffer than cables **122** and **124** such that they are not interchangeable. Thus, lighting unit **130** may be disposed above base **120** and third cable connector **125** may be disposed on a top of base **120** (e.g., the portion of the base disposed farthest from the ground or floor and closest to the ceiling).

(38) Another spaced lighting arrangement **202** for being spaced from the ground is shown, for example, in FIGS. **4** and **5**. A plurality of spaced lighting arrangements **202** (e.g., arranged in series) may be used in place of spaced lighting arrangements **102** shown in FIG. **1**. In some embodiments, a combination of spaced lighting arrangements **102** and spaced lighting arrangements **202** may be used in a single system **100**. Thus, lighting units may be installed to be spaced from the ceiling and other lighting units may be installed to be spaced from the ground in the same system **100**. This concept illustrates the complete modularity of system **100**.

(39) Spaced lighting arrangement **202** may comprise a base **220**, a lighting unit **230**, and a cable **224** connecting lighting unit **230** and base **220**. In some embodiments, base **220** comprises three cable connectors, including a first cable connector **221**, a second cable connector **223**, and a third cable connector **225**. Each cable connector **221**, **223**, **225** is configured to connect to a cable **108**, such as cable **224**. For example, cable connectors **221**, **223**, **225** may be threaded or have some other mating feature corresponding to an end of cables **108**. Cable connectors **221**, **223**, **225** may have any of the features of the cable connectors disclosed in U.S. application Ser. No. 16/929,668, filed Jul. 15, 2020, U.S. application Ser. No. 17/728,851, filed Apr. 25, 2022, and U.S. application Ser. No. 17/867,626, filed Jul. 18, 2022.

(40) In some embodiments, a first cable (such as cable **122** or a similar cable) is connected to first cable connector **221** (see FIG. **1** for the cable **122** and FIGS. **4** and **5** for the cable connector **221**). First cable **122** may connect base **220** to a first adjacent base **220** disposed upstream in a series of bases **220**. In some embodiments, a second cable (such as cable **126** or a similar cable) is connected to second cable connector **223** (see FIG. **1** for the cable **126** and FIGS. **4** and **5** for the cable connector **223**). Second cable **126** may connect base **220** to a second adjacent base **220** disposed downstream in a series of bases **220**. In some embodiments, a third cable **224** is connected to third cable connector **225** (see FIGS. **4-5**). Third cable **224** may connect base **220** to lighting unit **230**. In some embodiments, cables **122**, **224**, **126** are interchangeable such that they are configured to connect to any of first cable connector **221**, second cable connector **223**, and third cable connector **225**. In some embodiments, cable **224** may be stiffer than cables **122** and **126**. Thus, in some embodiments, cable **224** may not be interchangeable with cables **122** or **126**, as discussed above with respect to cable **107** and cables **103** and **105**.

(41) In some embodiments, first cable connector **221** and second cable connector **223** are disposed on opposite sides of base **220** from each other, as shown in FIGS. **4** and **5**. First cable connector **221** and second cable connector **223** may be aligned with each other such that they share a common axis. In other embodiments, first cable connector **221** and second cable connector **223** are disposed at angles with respect to each other. For example, first cable connector **221** and second cable connector **223** may be disposed at right angles to each other (i.e., 90-degree angles), which may facilitate a change in direction for a string of spaced lighting arrangements **202**. In some

embodiments, base **220** may have additional cable connectors, allowing a user to select which cable connector to use when assembling system **100**.

(42) In some embodiments, third cable connector **225** is disposed at a right angle (i.e., 90-degree angle) with respect to first cable connector **221** and second cable connector **223**. First cable connector **221**, second cable connector **223**, and third cable connector **225** may have their axes in a single plane. In some embodiments, third cable connector **225** is disposed on a top of the base **220** (e.g., the portion of the base disposed farthest from the ground or floor and closest to the ceiling), as shown in FIGS. **4** and **5**. Lighting unit **230** may be disposed above base **220**. With the first cable connector **221**, second cable connector **223**, and third cable connector **225** arranged as shown in FIGS. **4** and **5**, base **220** may be a T-shaped connection (e.g., an inverted T-shaped connection) between cables **122**, **224**, **126**.

(43) In some embodiments, cable **224** that connects lighting unit **230** to base **220** (via third cable connector **225**) comprises two ends **228**. One end **228** is connected to third cable connector **225** and the other end **228** is connected to a cable connector **235** (see FIG. **5**) of lighting unit **230**. Ends **228** may be the same, allowing cable **224** to be installed in either direction, furthering the modular nature of system **100**. Ends **228** may include threads or some other mating feature to correspond with cable connectors **225** and **235** (and other cable connectors of system **100**). To facilitate this, cable connectors **221**, **223**, **225**, **235** may each have the same shape and size. A gap is shown in cable **224** to represent that cables **224** of different lengths may be used as the cable **224** that connects a base **220** and a lighting unit **230**.

(44) As shown in FIG. **5**, base **220** may comprise a base circuit board **240**. In some embodiments, base circuit board **240** is disposed within base **220** (e.g., within a housing of base **220**). Base **220** may provide electrical connections between cable connectors **221**, **223**, **225** and base circuit board **240** so that power and communications from control unit **104** will reach base circuit board **240** and be able to be transmitted on to either lighting unit **230** or the next base **220**. In some embodiments, a microcontroller **242** is mounted on base circuit board **240**.

(45) As also shown in FIG. **5**, lighting unit **230** may comprise a lighting unit circuit board **250**. In some embodiments, lighting unit circuit board **250** is disposed within lighting unit **230** (e.g., within a housing of lighting unit **230**). Lighting unit **230** may provide electrical connections between cable connector **235** and lighting unit circuit board **250** so that power will reach lighting unit circuit board **250**. In some embodiments, one or more light emitting diodes **252** are mounted on lighting unit circuit board **250**.

(46) Microcontroller **242** may be configured to control light emitting diodes **252**. For example, microcontroller **242** may receive communication from control unit **104** and, based on that communication, light up light emitting diodes **252** so that lighting unit **230** emits light according to the communication from control unit **104**. By placing microcontroller **242** in base **220** (rather than in lighting unit **230**), less power can be transmitted along cable **224** than is transmitted along cables **122**, **126**. For example, power to light up the light emitting diodes **252** is all that is needed. Additional power to run the microcontroller **242** and send more complex commands is not needed through cable **224**. Thus, in some embodiments, power transmitted along cable **224** from base **220** to lighting unit **230** is less than power transmitted along cable **122** from an adjacent upstream base **220** to that particular base **220**. The power transmitted along cable **224** from base **220** to lighting unit **230** may also be less than power transmitted along cable **126** from that particular base **220** to an adjacent downstream base **220**. Because less power is being transmitted along the cables **224** connecting a base **220** with a lighting unit **230** to provide power to the lighting unit **230**, the total distance of cables **108** along which full power is being transmitted is shorter, thus improving the power supply to lighting units **230** farther away from the power source (e.g., control unit **104**).

(47) Because less power is being transmitted along cable **224**, cable **224** may be different than cables **122** and **126** such that cables **224** are not interchangeable with cables **122** and **126**, as discussed above for cable **107** with respect to cables **103** and **105**. For example, cable **224** may

have a smaller diameter and/or contain fewer wires. In some embodiments, cable **224** may have a larger diameter and/or be stiffer than cables **122** and **126**. However, in other embodiments, cable **224** is interchangeable with cables **122** and **126** (e.g., same diameter, same number of wires, same stiffness) to have greater modularity and simplify the setup of system **100**. Having different lengths does not make the cables non-interchangeable.

(48) As mentioned above, lighting system **100** may be used to space lights from the ceiling or the ground. For example, base **220** may be coupled to the ground with lighting unit **230** spaced from the ground (e.g., supported by a stiffness of cable **224**). In some embodiments, cable **124** is stiffer than cables **122** and **124** such that they are not interchangeable. In some embodiments, base **220** may be coupled to the ground via bolts, screws, or other fasteners. In some embodiments, base **220** may be coupled to the ground via adhesives. Base **220** may be coupled to the ground via other fastening means (e.g., hook and loop fasteners, etc.). Base **220** may be coupled to the ground via a combination of the disclosed techniques, or any other technique. In some embodiments, base **220** itself may include a component configured to couple to the ground. For example, as shown in FIGS. **4** and **5**, base **220** comprises a protruding structure **260**. Protruding structure **260** may be configured to extend into the ground to hold base **220** in place. In some embodiments, protruding structure **260** comprises one or more fins **262** to facilitate insertion into the ground. For example, protruding structure **260** may comprise four fins **262** at right angles with respect to each other, forming a “cross” shape. In some embodiments, fins **262** are larger near the rest of base **220** and gradually get smaller moving away from base **220** until they reach a point, allowing for easier insertion into the ground.

(49) In some embodiments, base **220** may instead be coupled to a ceiling with any of the techniques described above, or others. Thus, lighting unit **230** may be disposed below base **220** and third cable connector **225** may be disposed on a bottom of base **220** (e.g., the portion of the base disposed closest to the ground or floor and farthest from the ceiling).

(50) Where the above examples, embodiments and implementations reference examples, it should be understood by those of ordinary skill in the art that other modular lighting systems and lighting units could be intermixed or substituted with those provided. In places where the description above refers to particular embodiments of systems and methods of modular lighting, it should be readily apparent that a number of modifications may be made without departing from the spirit thereof and that these embodiments and implementations may be applied to other lighting technologies as well. Accordingly, the disclosed subject matter is intended to embrace all such alterations, modifications and variations that fall within the spirit and scope of the disclosure and the knowledge of one of ordinary skill in the art.

Claims

1. A modular lighting system comprising: a lighting unit comprising a lighting unit circuit board and one or more light emitting diodes mounted on the lighting unit circuit board; a base comprising three cable connectors, a base circuit board, and a microcontroller mounted on the base circuit board; a first cable connected to a first cable connector of the three cable connectors and connecting the base to a first adjacent base in a series of bases, the first adjacent base disposed upstream from the base; a second cable connected to a second cable connector of the three cable connectors and connecting the base to a second adjacent base in the series of bases, the second adjacent base disposed downstream from the base; and a third cable connected to a third cable connector of the three cable connectors and connecting the base to the lighting unit; wherein the first cable, the second cable, and the third cable are interchangeable such that the first cable, the second cable, and the third cable are configured to connect to any of the first cable connector, the second cable connector, and the third cable connector and configured to connect to any of the lighting unit, the first adjacent base, and the second adjacent base.

2. The modular lighting system of claim 1, wherein the base is configured to connect to a ceiling.
3. The modular lighting system of claim 2, wherein the lighting unit is disposed below and hanging from the base.
4. A modular lighting system comprising: a control unit; a plurality of bases arranged in series, wherein each base of the plurality of bases is configured to connect to a ceiling, and wherein a first base of the plurality of bases is connected to the control unit; a plurality of lighting units, each lighting unit connected to one of the plurality of bases; and a plurality of cables comprising a first cable connecting the control unit and the first base of the plurality of bases, a set of base cables, wherein each base cable connects two bases of the plurality of bases, and a set of lighting unit cables, wherein each lighting unit cable connects one of the plurality of lighting units to one of the plurality of bases such that each lighting unit is disposed below and hanging from one of the plurality of bases; wherein an amount of power transmitted along the lighting unit cables is less than an amount of power transmitted along the first cable and the base cables.
5. The modular lighting system of claim 4, wherein the first cable, the base cables, and the lighting unit cables are interchangeable such that each of the first cable, the base cables, and the lighting unit cables is configured to connect to the control unit, any base of the plurality of bases, and any lighting unit of the plurality of lighting units.
6. The modular lighting system of claim 4, wherein the first cable and the base cables are interchangeable.
7. The modular lighting system of claim 4, wherein the lighting unit cables are different than the first cable and the base cables such that the lighting unit cables are not interchangeable with the first cable or the base cables.
8. The modular lighting system of claim 7, wherein the lighting unit cables are stiffer than the first cable and the base cables.
9. The modular lighting system of claim 7, wherein a diameter of the lighting unit cables is different than a diameter of the first cable and a diameter of the base cables.
10. The modular lighting system of claim 9, wherein the diameter of the lighting unit cables is less than the diameter of the first cable and the diameter of the base cables.
11. The modular lighting system of claim 4, wherein each base comprises a base circuit board and a microcontroller mounted on the base circuit board, and wherein each lighting unit comprises a lighting unit circuit board and one or more light emitting diodes mounted on the lighting unit circuit board.